Lab: Derivation Exercise — Generative Models and Variational Inference

Objective

Derive the variational inference update equations for a concrete generative model, connecting the abstract mathematics to explicit computations.

Part 1: Hidden Markov Model Specification

Goal: Write out a complete generative model.

Consider a Hidden Markov Model with:

- $K = 3$ hidden states ($s \in \{1, 2, 3\}$)
- Observation model: $p(o_t \mid s_t = k) = N(o_t; \mu_k, \sigma_k^2)$ with known parameters
- Transition model: $p(s_t \mid s_{t-1})$ given by a 3×3 matrix **A** with rows summing to 1
- Prior: $p(s_1)$ = categorical with probabilities [0.5, 0.3, 0.2]

Task: Write the complete joint distribution $p(o_{1:T}, s_{1:T})$ for $T = 3$ timesteps. Expand the product notation fully.

Write your response here

Part 2: Variational Free Energy for the HMM

Goal: Derive the free energy for this model.

Let $q(s_{1:T}) = \prod_t q(s_t)$ be the mean-field recognition density, where each $q(s_t)$ is a categorical distribution with parameters (probabilities) $r_t = [r_{t1}, r_{t2}, r_{t3}]$.

Task: Write the variational free energy:

$F = E\_q[\ln q(s_{1:T})] - E\_q[\ln p(o_{1:T}, s_{1:T})]$

Expand each term. The first term is the negative entropy of q. The second term involves the log-likelihood and log-transition probabilities.

Write your response here

Part 3: Optimal Update Equations

Goal: Derive the fixed-point equations for the sufficient statistics.

Taking the derivative $\partial F/\partial r_{\boxed{?}\boxed{?}} = 0$ (with the constraint $\sum_{\boxed{?}} r_{\boxed{?}\boxed{?}} = 1$), derive the optimal $q^*(s_{\boxed{?}})$:

$$\ln q^*(s_{\boxed{?}} = k) \propto \ln p(o_{\boxed{?}} \mid s_{\boxed{?}} = k) + E\_\{q(s_{\boxed{?}\_1})\}[\ln p(s_{\boxed{?}} = k \mid s_{\boxed{?}\_1})] + E\_\{q(s_{\boxed{?}+1})\}[\ln p(s_{\boxed{?}+1} \mid s_{\boxed{?}} = k)]$$

Show all steps. Explain why the update depends on messages from both the past ($s_{\boxed{?}-1}$) and the future ($s_{\boxed{?}+1}$).

Write your response here

Part 4: Gaussian State-Space Model

Goal: Repeat the derivation for a continuous model.

Consider a linear Gaussian state-space model:

- $p(s_{\boxed{?}} \mid s_{\boxed{?}-1}) = N(s_{\boxed{?}}; A \cdot s_{\boxed{?}-1}, Q)$ — state dynamics
- $p(o_{\boxed{?}} \mid s_{\boxed{?}}) = N(o_{\boxed{?}}; C \cdot s_{\boxed{?}}, R)$ — observation model
- $p(s_1) = N(s_1; \mu_0, \Sigma_0)$ — prior

With recognition density $q(s_{\boxed{?}}) = N(s_{\boxed{?}}; m_{\boxed{?}}, V_{\boxed{?}})$.

Task: Derive the update equations for the sufficient statistics $m_{\boxed{?}}$ (mean) and $V_{\boxed{?}}$ (covariance) by minimizing F. Show that these correspond to Kalman filter/smoother equations.

Write your response here

Part 5: Synthesis

In 200 words, explain the common mathematical structure across Parts 2-4: in both discrete and continuous cases, the optimal recognition density arises from minimizing free energy, and the update equations combine likelihood (data-driven) terms with transition (model-driven) terms.

Write your response here

Lab Summary

Part	Skill Developed	Mathematical Result
1	Model specification	Writing the HMM joint distribution
2	Free energy derivation	Expanding F for mean-field HMM
3	Variational calculus	Deriving optimal update equations
4	Continuous extension	Gaussian state-space / Kalman filtering
5	Conceptual synthesis	Unifying discrete and continuous cases